

LEC 1
 intro
 linear equations
 linearly independent

$\mathbb{R}^n$ is the set of all n -dimensional vectors with real-valued entries
 $\mathbb{C}^n$ complex-valued
 $\mathbb{R}^{m \times n}$ is the set of matrices with real values and size $m \times n$
 $\mathbb{C}^{m \times n}$ complex

row picture $Ax = b$ $\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

col picture $x_1 \cdot \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} + x_2 \cdot \begin{bmatrix} A_{12} \\ A_{22} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

matrix-vector product can also be viewed as taking a linear comb of cols of the matrix

$$A \cdot B = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \begin{bmatrix} x & y & z \\ u & v & w \end{bmatrix} = \begin{bmatrix} col\ 1 & col\ 2 & col\ 3 \end{bmatrix}$$

$col\ 1 = A \cdot \begin{bmatrix} x \\ u \end{bmatrix} = x \cdot \begin{bmatrix} a \\ c \\ e \end{bmatrix} + u \cdot \begin{bmatrix} b \\ d \\ f \end{bmatrix}$

inner product $\langle x, y \rangle = x^T y$
 $x^T y = 0 \Rightarrow x$ and y are perpendicular
 in ML classification, we evaluate $\sqrt{x_{test}}$ $> / < 0$
 v is a vector perpendicular to the decision boundary

linearly independent, span $\mathbb{R}^2$: linear combs over entire $\mathbb{R}^2$

linear equations $\begin{cases} \text{single solution} \rightarrow \begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}, x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} -2 \\ 1 \end{bmatrix}$
 no solution
 $\text{infinite solution} \} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \text{ span } \mathbb{R}^2$

if $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$ holds only for $a_i = 0 \ \forall i \Rightarrow$ linearly independent
 otherwise, say $a_2 \neq 0$, $v_2 = -\frac{1}{a_2} \sum_{i \neq 2} a_i v_i \Rightarrow$ linearly dependent

LEC 2
 Gaussian elimination
 matrix multiplication
 matrix inverse
 length & angle of vector

$\left[\begin{array}{ccc|c} 1 & -1 & 3 & 9 \\ 0 & 1 & -14 & -27 \\ 0 & 0 & 1 & 2 \end{array} \right] \xrightarrow{-14} \xrightarrow{-3} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right] \xrightarrow{1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$

row echelon form reduced row echelon form (RREF)
same as back substitution

application of matrix multiplication: counting walks on graphs

a walk is a sequence of vertices connected by edges. repeats are ok
 e.g. a-b-d-c-b is a walk of length 4
 task: count # walks of a particular length

Def: for a given graph with n vertices, the adjacency matrix A is an $n \times n$ matrix

$A_{ij} = 1$ if edge between i & j exist and 0 otherwise

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$ $A^2 = A \cdot A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$

$a \ b \ c \ d$ $a \ b \ c \ d$

entry (1,1) in A^2 is # walks of the form a-x-a. (only a-b-a)
 entry (2,2) b-x-b (b-a-b, b-c-b, b-d-b)
 $(A^2)_{ik} = \sum_{j=1}^n A_{ij} A_{jk}$

walks of length 10: A^{10}
 all even-length walks: $A^2 + A^4 + A^6 + \dots$
 Katz centrality (early version of page rank): $B = \sum_{k=0}^{\infty} \alpha^k A^k \quad \alpha \in (0,1)$
 $C(i) = \sum_{j=1}^n B_{ji} \rightarrow$ used to rank nodes in social network by influence

commutativity $A \cdot B \neq B \cdot A$
 associativity $A(BC) = (AB)C$
 distributivity $A(B+C) = AB + AC$
 $(A+B)C = AC + BC$

Identity $I_{n \times n}$. $A_{m \times n} \cdot I_{n \times n} = A_{m \times n}$
 $I_{n \times n} \cdot A_{n \times m} = A_{n \times m}$

A^{-L} is left inverse of A iff $A^{-L}A = I$

A^{-R} right $AA^{-R} = I$

Fact: when A is square. $A^{-L}A = AA^{-R} = I$. $A^{-L} = A^{-R}$

proof: $A^{-L} = A^{-L}I = A^{-L}(A \cdot A^{-R}) = (A^{-L}A)A^{-R} = I \cdot A^{-R} = A^{-R}$

thus for square matrix. we denote inverse as A^{-1} (no left or right). $A^{-1}A = AA^{-1} = I$

$$Ax = b \quad A^{-L}A x = A^{-L}b \quad x = A^{-L}b$$

$$\underbrace{E_3 \cdot E_2 \cdot E_1}_{E=A^{-1}} \cdot A = I \quad [A | I] \rightarrow [EA=I | EI=A^{-1}]$$

$$E=A^{-1}$$

RREF

change A to RREF

other methods for finding inverses

① Cramer's rule (for low-dim): $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

② Eigendecomposition

$$I^{-1} = I \quad (I \cdot I = I)$$

a way to calculate matrix inverse

$$u \cdot x^{(i)} = e_i = \begin{bmatrix} 1 \\ 0 \\ \vdots \end{bmatrix} \quad u^{-1} = \begin{bmatrix} x^{(1)} & x^{(2)} & \dots & x^{(n)} \end{bmatrix}$$

e_2
 e_3

diagonal matrix: square. non-zero entries only on diagonal

$$\begin{bmatrix} d_1 & 0 \\ 0 & d_2 \\ \vdots & \vdots \\ 0 & \vdots & d_n \end{bmatrix} \begin{bmatrix} u_1 & 0 \\ 0 & u_2 \\ \vdots & \vdots \\ 0 & \vdots & u_n \end{bmatrix} = I$$

$d_1 u_1 = 1$
 $d_2 u_2 = 1$
 $\dots$
 $d_n u_n = 1$

if $d_i \neq 0 \forall i$. $A^{-1} = \text{diag}(\frac{1}{d_1}, \frac{1}{d_2}, \dots, \frac{1}{d_n})$

def. Euclidean length of vector / L_2 -norm $\|v\| = \sqrt{\sum v_i^2} = \sqrt{\langle v, v \rangle} = \sqrt{v^T v}$

$$\text{angle } \theta = \cos^{-1} \frac{\langle v, w \rangle}{\|v\| \|w\|}$$

LEC 3

orthogonal matrix

(permutation, rotation, reflection)

projection

vector space & subspace

orthogonal matrix Q : preserve vector inner product upon multiplication: $\forall v, w, \langle Qv, Qw \rangle = \langle v, w \rangle$

equivalently: preserve length & angle: $\|Qv\| = \|v\|$

permutation: consider a map

$$\pi \begin{matrix} a & b & c & d \\ \searrow & \searrow & \searrow & \searrow \\ d & a & b & c \end{matrix} \quad \begin{matrix} \text{pre-image} \\ \text{image} \end{matrix} \quad \begin{matrix} \pi(1)=2 & \pi(3)=4 \\ \pi(2)=3 & \pi(4)=1 \end{matrix}$$

① same size: $|\text{pre-image}| = |\text{image}|$

② each element in pre-image appears exactly once in image

$$\text{e.g. } \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow \begin{bmatrix} z \\ x \\ y \end{bmatrix}$$

permutation matrix: a sparse matrix with a single 1 in each row & each column

$\Rightarrow$ all rows are distinct, all columns are distinct

Q: if A and B are permutation matrix. what can we say about AB ?

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} z \\ x \\ y \end{bmatrix} \quad B \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ z \\ y \end{bmatrix}$$

$$\Rightarrow AB \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ z \\ x \end{bmatrix} \quad AB \text{ is also a permutation matrix}$$

permutation matrix does not change length & angle $\|v\| = \|Av\|$. $O_{v,w} = O_{Av, Aw}$

$\Rightarrow$ permutation matrix is orthogonal matrix

transpose of permutation matrix is also permutation matrix.

For permutation matrix $A \cdot B$. $A \cdot A^T = I \Rightarrow A^T = A^{-1}$

$$(A \cdot B)^T = (AB)^T = B^T A^T$$

in general, for any matrix A, B

$$(A \cdot B)^T = B^T A^T$$

$$(A+B)^T = A^T + B^T$$

$$(A^{-1})^T = (A^T)^{-1} \quad \text{proof: } A^T \cdot (A^{-1})^T = (A^{-1} A)^T = I^T = I$$

Rotation e.g.

$$x \rightarrow y \quad y \rightarrow -x$$

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -x \end{bmatrix}$$

rotation by angle θ : $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \cos \theta + y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix}$

after rotation, vector length & angle doesn't change $\Rightarrow$ rotation matrix are orthogonal matrix

$$(R_\theta)^{-1} = R_{(-\theta)} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = (R_\theta)^T$$

$$R_\theta \cdot R_{(-\theta)} = I \quad (R_\theta)^{-1} = (R_\theta)^T$$

$\Rightarrow$ orthogonal matrix $Q \Leftrightarrow Q^T Q = I$

$$(\Rightarrow): \langle Qv, Qw \rangle = \langle v, w \rangle \quad \forall v, w$$

$$(\Leftarrow): \langle Qv, Qw \rangle = v^T Q^T Q w = v^T I w = v^T w = \langle v, w \rangle$$

$$v^T Q^T Q w = v^T w$$

$$\text{set } v^T = [0 \dots 0 \overset{\text{ith}}{1} 0 \dots 0]$$

$$w^T = [0 \dots 0 \overset{\text{kth}}{1} 0 \dots 0]$$

$$(Q^T Q)_{jk} = I_{jk} \quad Q^T Q = I$$

$$\begin{bmatrix} -q_1^T \\ -q_2^T \\ \vdots \\ -q_n^T \end{bmatrix} \begin{bmatrix} | & | & \dots & | \\ q_1 & q_2 & \dots & q_n \\ | & | & \dots & | \end{bmatrix} = I$$

$$\langle q_i, q_j \rangle = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{else} \end{cases}$$

cols of Q are mutually orthogonal, unit vectors

other orthogonal matrix

reflection $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

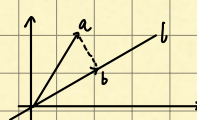
in general, reflection $w = I - 2uu^T$ (u is unit vector)

$$Uu = -u; \quad uw = v, \quad \forall v \perp u.$$

hadamard transform $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

haar wavelet transform, discrete Fourier transform

Projections



map a to b

def: $\vec{b}$ span l , closest to a

idempotent: $P^2 = P$, project 2 times same as 1 time

simpler case: l : x -axis

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \rightarrow \begin{bmatrix} v_1 \\ 0 \end{bmatrix}$$

l : y -axis

$$P = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

more generally: $P = R_\theta \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} R_{-\theta}$
 rotate project to undo rotate
 x -axis

$$P = A \cdot B \cdot C \Rightarrow P \text{ is not invertible}$$

application: dimension reduction

Vector Space

Def: A vector space over $\mathbb{R}$ is a set V of vectors with rules: ① addition, $\forall v, w \in V, v + w \in V$

e.g. ① set of all n -component vectors, $\mathbb{R}^n$

② scalar multiplication, $\forall v \in V, \forall d \in \mathbb{R}, dv \in V$

② set of all $m \times n$ matrices ($A+B, \alpha A$ are still $m \times n$)

③ set of all degree $\leq n$ polynomials $p(x) = G_0 + G_1 x + G_2 x^2 + \dots + G_n x^n$

④ set of fns over interval domain $[0, 1] \rightarrow$ infinite dimension

Subspace.

Def: Given a vector space V . a subset $S \subseteq V$ is a subspace iff $\begin{cases} \forall v, w \in S. v+w \in S \\ \forall \alpha \in \mathbb{R}. v \in S. \alpha v \in S \end{cases}$

e.g. subspaces of $\mathbb{R}^2$: $\mathbb{R}^2, \mathbb{R}^1, \{0\}$

any line passing through origin

LEC 4 vector space: S_1 and S_2 are 2 subspaces of vector space V

col/row space range. $\Rightarrow S_1 \cup S_2$ is not necessarily a subspace. counterexample: 2 lines in a plane

null space $S_1 \cap S_2$ is a subspace

polynomial fitting

rank & basis

$A_{m \times n} \Rightarrow n$ m-dim cols span $C(A) \subseteq \mathbb{R}^m$

$\{x: Ax=0\}$ $N(A) \subseteq \mathbb{R}^n$

m n-dim rows span $C(A^T) \subseteq \mathbb{R}^n$

For square matrix $A_{n \times n}$. A is invertible iff $C(A) = \mathbb{R}^n$

$\{x: Ax=c\}$ for $c \neq 0$ is not a subspace.

$Av=c$ $Aw=c$ $A(v+w)=2c \neq c$

suppose $\exists y \neq 0$ s.t. $y \in N(A)$ for some square matrix A . A is not invertible

$A \cdot y = 0$.

suppose A is invertible. we can solve $x = A^{-1}b$ uniquely

$A(x+y) = Ax + Ay = b + 0 = b$

$x, x+y$ are solutions. contradiction

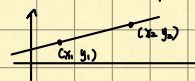
For square matrix, if non zero vector exist in $N(A) \Rightarrow C(A) \neq \mathbb{R}^n$

Application: (Over) Fitting data with polynomials

suppose you have 5 dpts. $(x_1, y_1) (x_2, y_2) \dots (x_5, y_5)$

goal: fit a polynomial to data s.t. upon seeing a new point x_t . we can predict y_t

simpler case: 2 dpts. $y = ax + b$

 unknown: a, b
Given: $(x_1, y_1) (x_2, y_2)$

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \end{bmatrix} \begin{bmatrix} b \\ a \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

back to 5 pts. to fit perfectly. $y = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4$

$$\begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & x_1^4 \\ 1 & x_2 & x_2^2 & x_2^3 & x_2^4 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_5 & x_5^2 & x_5^3 & x_5^4 \end{bmatrix} \begin{bmatrix} C_0 \\ C_1 \\ C_2 \\ C_3 \\ C_4 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_5 \end{bmatrix}$$

A

Solving for C_i is what we do when fitting the polynomial

if A is invertible. we get perfect fit.

in practice: degree of polynomial $\ll$ data size

rank / col rank: $\dim C(A)$

row rank: $\dim C(A^T)$

Basis set

a set B is a basis for a vector space V

iff $\text{span}(B) = V$

$\ominus B$ is linearly independent set

orthogonal basis: (e.g. cols of I_n). $\|e_k\|=1$. $e_j^T e_k = 0 \forall j \neq k$ (orthogonal)

LEC 5 Basis

orthogonality

orthogonal complement

Gram-Schmidt

Basis for all 3×3 symmetric matrices. $\dim = 3 = \frac{n(n+1)}{2}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

pairwise orthogonal vectors are linearly independent

if V and W are orthogonal spaces. $V, W \subseteq \mathbb{R}^n$. then $\dim V + \dim W \leq n$
 orthogonal complement of a subspace must be a subspace

e.g. $\dim V + \dim V^\perp = 2$

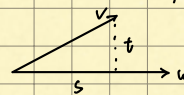
(1) $V = \mathbb{R}^2 \xrightarrow{\text{line}} V^\perp = \{\vec{0}\}$

(2) $V = \text{line} \rightarrow V^\perp = \text{perpendicular line}$

$V = \text{span} \left\{ \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}_{v_2} \right\}$. what is $V^\perp$?

$\Rightarrow \begin{cases} w^T v_1 = 0 \\ w^T v_2 = 0 \end{cases} \Rightarrow \begin{cases} w_1 + 2w_2 + 3w_3 = 0 \\ w_1 + w_3 = 0 \end{cases} \Rightarrow V^\perp = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$
 $w_1 = x, w_2 = x, w_3 = -x$

Given vector w with $\|w\|=1$. projection of v on w is $\text{proj}_w v = \underbrace{(v^T w)}_{\text{scalar}} w = \underbrace{(ww^T)}_{\text{projection matrix}} v$



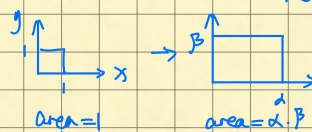
LEC 6
 Linear trans
 Det
 proof of fundamental
 results

linear trans $L: V \rightarrow W$. $\forall \alpha, \beta \in \mathbb{R}, \forall x_1, x_2 \in V$. $L(\alpha x_1 + \beta x_2) = \alpha L(x_1) + \beta L(x_2) \in W$
 can always be described via a matrix. $L(x) = Ax$

determinant: how the volume of a set changes under L

$S \xrightarrow{L} L(S)$
 $\frac{\text{Vol}(L(S))}{\text{Vol}(S)}$

e.g. $A = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}$ $A \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \alpha x \\ \beta y \end{bmatrix} \Rightarrow \det A = \alpha \cdot \beta$



e.g. $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$ $\det A = -1$

e.g. rotation $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ $\det A = 1$

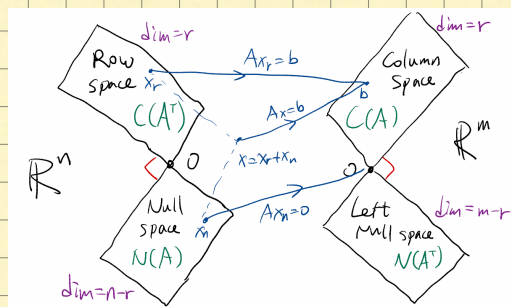
e.g. projection. not invertible. $\det A = 0$

Theorem 1: Let $V \subseteq \mathbb{R}^n$ be a subspace.

Any $x \in \mathbb{R}^n$ can be uniquely decomposed into $x = x_1 + x_2$ where $x_1 \in V, x_2 \in V^\perp$

Theorem 2: For any subspace $V \subseteq \mathbb{R}^n$. $\dim V + \dim V^\perp = n$

LEC 7
 Fundamental spaces
 rank-nullity theorem
 matrix inverse



Theorems (3) $N(A)^\perp = C(A^T)$

$C(A)^\perp = N(A^T)$

(4) $\dim C(A) + \dim N(A) = n$

② $\dim C(A) = \dim C(A^T) = r$
 Rank-Nullity: $\text{rank}(A) + \text{nullity}(A) = n$
 $\downarrow \quad \downarrow$
 $r \quad \dim N(A)$

find basis for 4 subspaces

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} \rightarrow R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

basis for row space: non-empty rows of R

$$C(A^T) = C(CR^T) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$$

basis for col space: pivot cols

$$C(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}$$

basis for null space: let $G = \begin{bmatrix} -F \\ I \end{bmatrix} \cdot RG = 0$

$$\underbrace{R}_{\text{invertible}} A \cdot \Rightarrow AG = 0$$

$$N(A) = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\} \rightarrow \text{ols of } G$$

basis for left null space: $E[A|I] = [R|E]$ to find E

$$EA = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} = R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$N(A^T) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

Inverse

left & right inverse may not be unique, and may not exist

left exist iff $r=n$
 right exist iff $r=m$

if A has both left inverse and right inverse, A must be square $r=m=n$. $A^{-L} = A^{-R} = A^{-1}$

One left inverse of A is $A^{-L} = (A^T A)^{-1} A^T$
 right $A^{-R} = A^T (A A^T)^{-1}$ proof: $A \cdot A^{-R} = A A^T (A A^T)^{-1} = I$

positive definite matrix: square symmetric matrix M . $x^T M x > 0 \quad \forall x \neq 0$
 posdef matrix is invertible

A has full col rank, $r=n$. $\Leftrightarrow A^T A$ is posdef, thus invertible $\Leftrightarrow A$ has left inverse

① $A^T A$ is symmetric

② $x^T A^T A x = \|Ax\|_2^2 > 0 \quad \forall x \neq 0$

$N(A) = \{0\} \Rightarrow Ax = 0$ only for $x=0$

LEC8

Sylvester's Criterion
 for posdef

Subspace projection

Pseudo-Inverse (optional)

A is square. A is invertible iff

① A has full col rank

② A has zero null space

③ A is left invertible

④ A has full row rank

⑤ A is right invertible

⑥ $Ax=b$ has exactly one solution $\forall b$ and it's given by $x = A^{-1}b = A^{-L}b$

$\Rightarrow A^{-1} = A^T (A A^T)^{-1} = (A^T A)^{-1} A^T$

$$(A^T)^{-1} = (A^{-1})^T \quad \text{proof: } (A^{-1})^T A^T = (A A^{-1})^T = I^T = I$$

AB must be same size square. $(AB)^{-1} = B^{-1}A^{-1}$

proof: let $C = AB$

$$B^{-1}A^{-1}C = B^{-1}A^{-1}AB = B^{-1}B = I$$

eg. $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ Find left inverse B

$$B_1 = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 1 \end{bmatrix} \quad B_2 = \begin{bmatrix} -1 & 1 & -1 \\ -2 & 1 & 0 \end{bmatrix}$$

$\alpha B_1 + \beta B_2 \quad \forall \alpha + \beta = 1$ is a left inverse

Sylvester's Criterion: Let A_k be the $k \times k$ upper left corner of A

A is posdef $\Leftrightarrow \det(A_k) > 0 \quad \forall k = 1, 2, \dots, n$

intuition: $n=2$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

posdef: $x^T A x > 0 \quad \forall x \neq 0$

$$a_{21} = a_{12}$$

$$\text{let } x = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} \quad x^T A x = x_1^2 a_{11} > 0 \Leftrightarrow a_{11} > 0$$

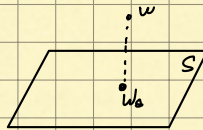
$$\begin{aligned} \text{let } x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad x^T A x &= a_{11} x_1^2 + 2a_{12} x_1 x_2 + a_{22} x_2^2 \\ &= a_{11} (x_1 + \frac{a_{12}}{a_{11}} x_2)^2 + \frac{1}{a_{11}} (a_{11} a_{22} - a_{12}^2) x_2^2 > 0 \\ &\quad \text{det } A > 0 \end{aligned}$$

Projection on subspace

notation: $w_s = \text{proj}_S(w)$

(a) $w_s \in S$

(b) $(w - w_s) \perp S$



Let $B = \{v_1, \dots, v_k\}$ be an orthonormal basis for S

$$\text{proj}_S(w) = \sum_{i=1}^k (v_i^T w) v_i$$

$$\text{verify (b)} \quad [w - \sum_{i=1}^k (v_i^T w) v_i]^T v_j = w^T v_j - \sum_{i=1}^k (v_i^T w) v_i^T v_j = w^T v_j - (v_j^T w) \underbrace{v_j^T v_j}_1 = 0 \quad \forall j$$

$$\text{projection matrix} \quad \text{proj}_S(w) = \sum_{i=1}^k (v_i^T w) v_i = \underbrace{\begin{bmatrix} v_1 & \dots & v_k \end{bmatrix}}_{P_S} w$$

Given any orthonormal basis $\{v_1, \dots, v_k\}$

let $V = \begin{bmatrix} | & & | \\ v_1 & \dots & v_k \\ | & & | \end{bmatrix} \quad n \times k \quad (\mathbb{R}^k \text{ is a subspace of } \mathbb{R}^n)$

$$V^T V = I_{k \times k}$$

$V V^T = P_S \quad n \times n$, project onto $\text{span}\{v_1, \dots, v_k\}$

$P^2 = P$. idempotent

$A \in \mathbb{R}^{n \times n}$. ① suppose full row rank $m=r$

given $x \in \mathbb{R}^n$. decompose $x = x_1 + x_2$. $x_1 \in N(A)$. $x_2 \in C(A^T)$

$$\downarrow \\ x_2 = A^T z \text{ for some } z$$

$$x_1 = x - x_2 = x - A^T z$$

$$A x_1 = 0 \Rightarrow A x - A A^T z = 0$$

$$A x = A A^T z$$

A has full row rank $\Leftrightarrow A A^T$ invertible

$$z = (A A^T)^{-1} A x$$

$$\Rightarrow x_2 = \underbrace{A^T (A A^T)^{-1} A}_{\text{ortho proj onto } C(A^T)} x \in C(A^T)$$

$$x_1 = \underbrace{[I - A^T (A A^T)^{-1} A]}_{\text{ortho proj onto } N(A)} x \in N(A)$$

$$P_{\text{row}} = A^T (A A^T)^{-1} A$$

$$P_{\text{col}} = I_{m \times m}$$

$$P^2 = P$$

$$A^+ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0 \quad A^+ x = 0 \quad \forall x$$

② suppose full col rank. decompose $X \in \mathbb{R}^m = x_1 \in C(A) + x_2 \in N(A^T)$

$$Q = A(A^T A)^{-1} A^T$$

ortho proj onto $C(A)$

$$Q^\perp = I - A(A^T A)^{-1} A^T$$

$$P_{col} = A(A^T A)^{-1} A^T$$

$$P_{row} = I_{mn}$$

$$\text{Generally: } P_{col} = AA^+ \quad P_{left null} = I - P_{col}$$

$$P_{row} = A^+ A \quad P_{null} = I - P_{row}$$

③ rank $A < m, n$

PseudoInverse / Moore-Penrose Inverse A^+ def:

one is I

$$① A A^+ A = A$$

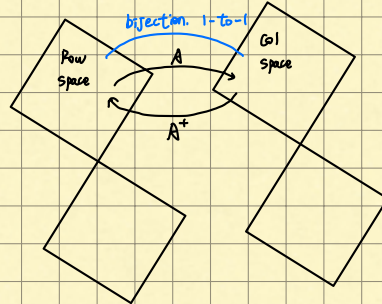
$$② A^+ A A^+ = A^+$$

$$③ (A A^+)^T = A A^+$$

$$④ (A^+ A)^T = A^+ A$$

$\Rightarrow$ generalized inverse (not unique)

uniquely define A^+



$$A^+ x = 0 \quad \forall x \in N(A^T)$$

A^+ always exist for any $m \times n$ matrix and is unique

e.g.

$$\text{if } A^{-1} \text{ exist, } A^+ = A^{-1}$$

$$\text{if } A^+ / A^{-R} \text{ exist, } A^+ = A^{-L} \text{ or } A^{-R}$$

$$\text{For any scalar } d, d^+ = \begin{cases} d^{-1} & \text{if } d \neq 0 \\ 0 & \text{if } d = 0 \end{cases}$$

$$\begin{bmatrix} 2 & -3 \\ 0 \end{bmatrix}^+ = \begin{bmatrix} \frac{1}{2} & -\frac{1}{3} \\ 0 \end{bmatrix}$$

If A is symmetric with $A = U D U^T$ where U is orthogonal and D is diagonal

$$\text{then } A^+ = U D^+ U^T$$

$$A^+ = (A^T A)^+ A^T = A^T (A A^T)^+$$

$$(AB)^+ \neq B^+ A^+ \quad \text{e.g. } A = \begin{bmatrix} 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$(AB)^+ = \begin{bmatrix} 1 \\ 1 \end{bmatrix}^+ = 1$$

$$B^+ A^+ = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{2}$$

$$A A^+ = \text{ortho proj onto } C(A)$$

$$A^+ A = \text{ortho proj onto } C(A^T)$$

$$A^+ = A^{-1} \text{ or canonical } A^{-L} / A^{-R} \text{ if they exist}$$

least square

$$\min_x \|Ax - b\| \quad x^+ = A^+ b$$

$$Ax = b$$

$$x = A^+ b$$

If unique sol, give that sol

If many sols, give min norm sol (wide)

If no sol, give min square sol (tall)

LEC 9 solve $Ax = b$

$$A x = b \quad \mathbb{R}^n \rightarrow \mathbb{R}^m$$

(a) no sol

(b) infinite many sol

(c) unique sol

$$(a) \Leftrightarrow b \notin C(A)$$

$$(b) \Leftrightarrow b \in C(A)$$

$$(c) \Leftrightarrow N(A) = \{0\} \quad \text{only possible when } n \leq m$$

$$(b) \Leftrightarrow N(A) \neq \{0\}. \quad x = x_{\text{particular}} + x_N$$

$$\exists \text{ unique sol } \forall b \in \mathbb{R} \Leftrightarrow A \text{ is invertible (square)} \quad x = A^{-1} b$$

$$\begin{bmatrix} \\ \end{bmatrix}$$

write underdetermined

(a) or (b)

$$\begin{bmatrix} \\ \\ \end{bmatrix}$$

tall overdetermined

typically no sol

but (a) (b) (c) are possible

$$\begin{matrix} m \times n & n \times 1 \\ A & x = b \end{matrix} \quad m > n.$$

$$A^T A \hat{x} = A^T b$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

can also view as left inverse $Ax = b$

A full col rank $A^{-L} Ax = A^{-L} b \quad A^{-L} = (A^T A)^{-1} A^T$

$$x^* = (A^T A)^{-1} A^T b$$

when full col rank and $m > n$. A^{-1} exist and is not unique
but x^* is unique

if ML.

$$\min_x \|Ax - b\|^2 + \lambda \|x\|^2 \quad \lambda \geq 0 \quad \text{regularization}$$

$$L(x) = \|Ax - b\|^2 = x^T A^T A x - 2x^T A^T b + b^T b + \lambda \|x\|^2$$

$$\frac{\partial L}{\partial x} = 2A^T A x - 2A^T b + 2\lambda x = 0$$

$$x = \underbrace{(A^T A + \lambda I)^{-1}}_{\text{nicer matrix guaranteed to be posdef \& invertible}} A^T b$$

if $m \leq n$. A full row rank

$$Ax = b$$

$$x = A^{-R} b$$

$$x^* = A^T (A A^T)^{-1} b \quad \text{least norm solution} \quad (\min \|x\|^2)$$

$$Ax = b \quad x = A^+ b$$

when $b \notin C(A)$ this x may not be exact ($Ax \neq b$). it is least square

when $N(A) \neq \{0\}$ this x may not be unique. it is least norm

General solution $x = A^+ b + \underbrace{(I - A^+ A)y}_{\text{project to } N(A)} \quad \begin{matrix} \text{infinite many sols} \\ y \in \mathbb{R}^n \text{ is arbitrary} \end{matrix}$

Midterm review

When row-reducing $[A \mid b]$, if you get a contradiction row:

$$[0 \ 0 \ \dots \ 0 \mid 1],$$

then the system has **no solution**.

- If no contradiction row appears, a solution exists.
- Unique vs infinitely many solutions depends on pivots vs free variables.

Positive definite (PD): quick facts

Definition (for symmetric M):

$$M \succ 0 \iff x^T M x > 0 \quad \forall x \neq 0.$$

High-yield facts:

- If $M \succ 0$, then all diagonal entries are positive: $M_{ii} > 0$.
- Identity to remember: $x^T (A A^T) x = \|A^T x\|_2^2$.
- If A is invertible, then $A A^T$ is PD (because $A^T x \neq 0$ when $x \neq 0$).

Projection onto a unit vector w ($\|w\|_2 = 1$):

$$\text{proj}_w(v) = (v^T w) w.$$

Matrix form (one-line consequence):

$$\text{proj}_w(v) = ww^T v \Rightarrow P = ww^T.$$

If w not unit-length, then

$$P = \frac{ww^T}{w^T w} \leftarrow \text{normalization}$$

Identification signatures: $P^2 = P$ (and for orthogonal projection: $P^T = P$).

$\uparrow$
hold for
non-perpendicular projection

$V - PV \perp S$. perpendicular projection

G-S. output is different for different orders of input vectors

Case	Shape	Rank	Solvable when	Solution
Left inverse	$m \geq n$	full column	$b \in \mathcal{R}(A)$	$x = A_L b$ (unique)
Right inverse	$m \leq n$	full row	all b	$x = A_R b + (I - A_R A) z$

$$Ax = b$$

project z into $N(A)$

$$A(A^R b) = b$$

thus $A^R b$ is a solution

$$A(x - A^R b) = Ax - b = 0$$

$$x - A^R b \in N(A)$$

LEC 10 QR decomposition

matrix norm $\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}$

property a) $\|Av\| \leq \|A\| \|v\|$

b) $\|AB\| \leq \|A\| \|B\|$

c) $\|A+B\| \leq \|A\| + \|B\|$ (triangle inequality)

For orthogonal matrix Q . $\|Q\| = \max_{x \neq 0} \frac{\|Qx\|}{\|x\|} = 1$

Cauchy - Schwarz Inequality

$$-1 \leq \frac{x^T y}{\|x\| \|y\|} \leq 1$$

cos π as 0

$$-\|x\| \|y\| \leq x^T y \leq \|x\| \|y\| \cdot \text{take} = \text{iff } x \text{ is parallel to } y$$

all matrix have QR decomposition

For overdetermined system

$$A_{m \times n} \text{ full col rank} \Leftrightarrow A \text{ left invertible} \quad \begin{matrix} n \times m & n \times n \\ L & A = I \end{matrix}$$

$$A \text{ can have } A = \begin{matrix} m \times m & m \times n \\ Q & R \\ \text{orthogonal} & \text{upper triangular} \end{matrix}$$

$$Ax = b$$

$$A^T A x = A^T b$$

$$R^T Q^T Q R x = R^T A^T b$$

$$R^T R x = R^T Q^T b$$

let R^T be L . lower-triangular matrix . if R is invertible . then so is L

$$L R x = L Q^T b \cdot \text{left multiply } L^{-1} \text{ both sides}$$

$$R x = Q^T b$$

$$\begin{bmatrix} \diagup \\ 0 \end{bmatrix} x = Q^T b$$

then back substitution

QR decomposition e.g.

$$A = \begin{bmatrix} 1 & -2 \\ 2 & -4 \\ 0 & 1 \end{bmatrix} \text{ is left invertible}$$

$$Q = \begin{bmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \\ q_{31} & q_{32} \end{bmatrix} \quad R = \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix}$$

First col of A

$$\begin{bmatrix} r_{11} & q_{11} \\ r_{21} & q_{12} \\ r_{31} & q_{13} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$Q^T Q = I \Rightarrow q_{11}^2 + q_{21}^2 + q_{31}^2 = 1$$

thus $q_{11} : q_{21} : q_{31} = 1 : 2 : 0, q_{31} = 0$

$$q_{11}^2 + q_{21}^2 = 1$$

$$q_{11} = \frac{1}{\sqrt{5}}, q_{21} = \frac{2}{\sqrt{5}}, q_{31} = 0$$

$$\begin{cases} \frac{1}{\sqrt{5}} r_{12} + q_{12} r_{22} = -2 \\ \frac{2}{\sqrt{5}} r_{12} + q_{22} r_{22} = -4 \\ q_{32} r_{22} = 1 \\ q_{12}^2 + q_{22}^2 + q_{32}^2 = 1 \\ \frac{1}{\sqrt{5}} q_{12} + \frac{2}{\sqrt{5}} q_{22} + q_{32} q_{32} = 0 \end{cases}$$

$$\text{let } Q = [q_1 q_2] \quad A = [a_1 a_2]$$

$$r_{12} q_1 + r_{22} q_2 = a_2 \quad \text{left multiply } q_1^T$$

$$r_{12} \frac{q_1^T q_1}{1} + r_{22} \frac{q_1^T q_2}{0} = q_1^T a_2$$

$$r_{12} = q_1^T a_2 = -2\sqrt{5}$$

$$q_{32} = 1 \quad r_{22} = 1 \quad q_{12} = 0 \quad q_{22} = 0$$

In general, we recursively build Q and R

$$\text{key setup} \quad \begin{bmatrix} a_1 & a_2 \end{bmatrix} = \begin{bmatrix} q_1 & q_2 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix}$$

$$\text{solve } q_1^T r_{11} \text{ first, which produce } R_{12} \quad q_1^T r_{11} = a_1 \quad q_1^T q_1 = 1 \Rightarrow r_{11} = \|a_1\|, q_1 = \frac{a_1}{\|a_1\|}$$

then do QR composition to solve $a_2 - R_{12} q_1$

$$Q^T Q = \begin{bmatrix} q_1^T \\ q_2^T \end{bmatrix} \begin{bmatrix} q_1 & q_2 \end{bmatrix} = \begin{bmatrix} q_1^T q_1 & q_1^T q_2 \\ q_2^T q_1 & q_2^T q_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & I_{n-1} \end{bmatrix}$$

$$\begin{bmatrix} a_1 & a_2 \end{bmatrix} = \begin{bmatrix} r_{11} & q_1 & q_1^T a_2 - q_1^T r_{12} \\ 0 & r_{22} \end{bmatrix}$$

$$a_2 = q_1 r_{12} + q_2 r_{22}$$

$$q_1^T a_2 = q_1^T q_1 r_{12} + q_1^T q_2 r_{22} = r_{12}$$

next, perform QR decomposition on

$$a_2 - q_1 r_{12} = a_2 - q_1 r_{12}$$

$$= a_2 - q_1 q_1^T a_2$$

$$= (I - q_1 q_1^T) a_2$$

project to $\text{span}\{q_1\}^\perp$

Connection to G-S numerical instability when a_1, a_2 are close

$$A = [a_1 a_2 \dots a_n] \xrightarrow{\text{G-S}} [w_1 w_2 \dots w_n] \quad \text{the recursive algorithm is more stable}$$

$$a_1 = w_1 (w_1^T a_1)$$

$$a_2 = w_1 (w_1^T a_2) + w_2 (w_2^T a_2)$$

$$a_k = \sum_{j=1}^k w_j (w_j^T a_k)$$

using $A = QR$ to solve $Ax = b$

if $b \notin \text{Col}(A)$, will get least square solution

in high-dim, do QR decomposition is cheaper than do A^{-1}

pos def: symmetric + $x^T A x > 0 \quad \forall x \neq 0$

pos semi def: symmetric + $x^T A x \geq 0 \quad \forall x$

If use Sylvester's criterion to test pos semi def, need to look at all principal minors instead of upper left ones

$$\text{e.g. test pos semi def} \quad \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$1 \times 1 \quad 2 > 0 \quad 2 > 0 \quad 1 > 0$$

$$2 \times 2 \quad \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} > 0 \quad \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} > 0 \quad \begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix}$$

$$3 \times 3 \quad \begin{vmatrix} 2 & -1 & 1 \\ -1 & 2 & 0 \\ 1 & 0 & 1 \end{vmatrix} > 0$$

σ_i is called singular value

$$V^T A V = V^T \Lambda V \quad \lambda_i = \sigma_i^2$$

LEC 11 SVD

$$\|Av\|^2 = \lambda \|v\|^2$$

$$\lambda = \frac{\|Av\|^2}{\|v\|^2} > 0. \text{ eigenvalues of } A^T A \text{ are non-negative}$$

any symmetric matrix have an orthonormal basis of eigenvectors

$$\text{let } v_1 = [v_{11} \dots v_{1n}]^T \quad v_2 = [v_{21} \dots v_{2n}]^T \quad S = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \quad U_1 = [u_1 \dots u_r] \quad U_2 = [u_{r+1} \dots u_m] \quad V = [V_1 | V_2] \quad U = [U_1 | U_2]$$

$$A^T A v_i = v_i \sigma_i^2$$

$$U_i = A v_i \sigma_i^{-1}$$

$$v_i^T A^T A v_i = \sigma_i^2$$

pick any $U_2 \in \mathbb{R}^{m \times (m-r)}$ s.t. U is orthogonal (through G-S)

$$S^{-1} v_i^T A^T A v_i S^{-1} = I$$

$$U^T A V = \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix} = \Sigma \quad A = [U_1 \ U_2] \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_1^T \\ V_2^T \end{bmatrix}$$

$$A V_2 = 0$$

$$A = U \Sigma V^T = \underbrace{U_1 S V_1^T}_{\text{compact SVD}} = \sum_{i=1}^r \sigma_i u_i v_i^T$$

$$\text{rank}(U) = \text{rank}(V) = \text{rank}(A) = r$$

SVD is not unique. the σ_j in S are unique

procedure ① eigenvalues of $A^T A$ $\lambda_i \Rightarrow \sigma_i = \sqrt{\lambda_i}$

② find v_i from $(A^T A - \lambda_i I) v = 0 \Rightarrow u_i = \frac{A v_i}{\sigma_i}$ for $\sigma_i > 0$

③ $\sigma_i = 0 \Rightarrow$ any u_i s.t. $u_j^T u_i = 0 \quad \forall j \in [0, i-1]$

$$\text{e.g. } A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{vmatrix} \lambda - 1 & 0 \\ 0 & \lambda \end{vmatrix} = 0 \quad \lambda = 1/0 \quad \sigma_1 = \sqrt{1} \quad \sigma_2 = 0$$

$$\sigma_1 = \sqrt{1}. \quad \lambda I - A^T A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} v_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} -1 \\ 0 \end{bmatrix} \text{ thus SVD not unique}$$

$$\sigma_2 = 0 \quad \lambda I - A^T A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ (or } \begin{bmatrix} 0 \\ -1 \end{bmatrix} \text{)}$$

$$u_1 = \frac{A v_1}{\sigma_1} = \begin{bmatrix} 1/\sqrt{1} \\ 0 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 1/\sqrt{1} \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ -1 \end{bmatrix} \dots$$

Preview: Pseudo inverse via SVD

$$A = U \Sigma V^T = U_i S V_i^T = \sum_{j=1}^r \sigma_j u_j v_j^T$$

$$\text{Then } A^+ = V \Sigma^+ U^T = V_i S^{-1} U_i^T = \sum_{j=1}^r \sigma_j^{-1} u_j v_j^T$$

LEC 12

$$\text{Matrix norm. } \|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \frac{\sqrt{x^T A^T A x}}{\sqrt{x^T x}} = \max_j \sqrt{\lambda_j} = \max_j \sigma_j \text{ (largest singular value of } A)$$

$$\text{low-rank approximation } A \approx A_b = \sum_{j=1}^b \sigma_j u_j v_j^T \quad \text{typically choose } b=1 \text{ or } b \ll \text{rank } A$$

$$A - A_b = \sum_{j=b+1}^r \sigma_j u_j v_j^T$$

largest singular value is σ_{b+1}

$$\|A - A_b\| = \sigma_{b+1} \ll \|A\| = \sigma_1$$

$$A v_i = \sum_{j=1}^r \sigma_j u_j v_j^T v_i = \sigma_i u_i$$

$$A^T u_i = \sigma_i v_i$$

if A is invertible

$$A^{-1} = (U \Sigma V^T)^{-1} = (V^T)^{-1} \Sigma^{-1} U^{-1} = V \Sigma^{-1} U^T$$

if A is not invertible

$$\text{Moore-Penrose Pseudoinverse } A^+ = \sum_{j=1}^r \sigma_j^{-1} v_j u_j^T$$

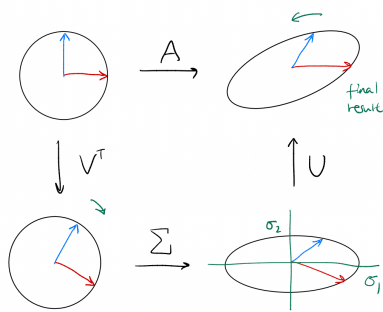
Intuition: $\exists x$ s.t. $Ax=b$, $b \notin \text{CCA}$ $\Rightarrow$ project onto CCA

$$AA^T = \left(\sum_{j=1}^r \sigma_j u_j v_j^T \right) \left(\sum_{k=1}^r \sigma_k u_k v_k^T \right) = \sum_{j=1}^r u_j u_j^T \Rightarrow \text{project onto } \text{CCA}$$

$$A^T A = \sum_{j=1}^r v_j v_j^T \Rightarrow \text{project onto } \text{CCA}^T$$

Visualization of SVD

$$A = U \Sigma V^T$$



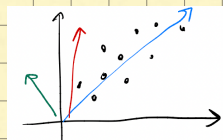
e.g. $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} I \begin{bmatrix} c & s \\ s & -c \end{bmatrix}$
 $c = \cos \theta, s = \sin \theta$

LEC 13 PCA Eigenvalue/vector diagonalization

Given $X_1, X_2, \dots, X_N \in \mathbb{R}^D$

want to reduce dimension to $M \ll D$

project to most informative M -dim subspace



let $X_1, X_2, \dots, X_N \in \mathbb{R}^D$ be data vectors. $X = [X_1 \dots X_N]_{D \times N}$

project to $M=1$ dim, project to u s.t. $\|u\|=1$

① find mean

sample mean $\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$

② center the data. for proper covmat

$$X^c = [X_1 - \bar{X} \dots X_N - \bar{X}]_{D \times N}$$

③ construct covmat.

$$S_{D \times D} = \frac{1}{N} X^c X^{cT}$$

④ principal direction.

find eigenvalue & vector of S . $\lambda_1 \geq \lambda_2 \geq \dots$
 $S u_1 = \lambda_1 u_1, \|u_1\|=1$

$D \times D$

⑤ project onto 1D u_1

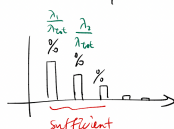
$$z_i = u_1^T (X_i - \bar{X}) \Rightarrow \text{a scalar. due to 1D}$$

project to M -dim

$\Rightarrow$ look for M largest eigenvalues of S

project to those M orthogonal eigenvectors

In practice, it is often the case that the first few dimensions carry most of the information. This is commonly captured via "Variance explained"



$$\lambda_{\text{tot}} = \sum_{i=1}^D \frac{\sigma_i^2}{N}$$

Fact 1: ① projected mean $u^T \bar{X}$

same $\langle c_2 \rangle$ mean of projected data $\frac{1}{N} \sum_{i=1}^N u^T X_i = \left(\frac{1}{N} \sum_{i=1}^N X_i^T \right) u = \bar{X}^T u$

Fact 2: let $z_i = u^T X_i, \bar{z} = u^T \bar{X}$. u is an 1D projection

PCA is max var in the projected dimension:

$$\Rightarrow \max_u \sum_{i=1}^N \frac{1}{N} (z_i - \bar{z})^2$$

scalar due to 1D

$$\text{var} = \sum_{i=1}^N \frac{1}{N} (z_i - \bar{z})^2 = u^T \left(\sum_{i=1}^N \frac{1}{N} (X_i - \bar{X})(X_i - \bar{X})^T \right) u = u^T S u$$

$\uparrow$
covmat of X^c

$$X^c = U \Sigma V^T, S = \frac{1}{N} U \Sigma \Sigma^T U^T$$

$$\Rightarrow \max_u u^T S u = \max_u \frac{1}{N} u^T U \Sigma \Sigma^T U^T u \text{ s.t. } \|u\|=1$$

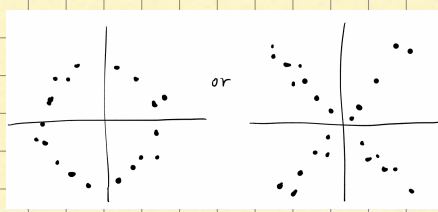
set $b = U^T u, u = U b$. U is orthogonal and norm preserving $\|u\|=1 \Rightarrow \|b\|=1$

$$\Rightarrow \max_b \frac{1}{N} b^T \Sigma \Sigma^T b$$

$$\Sigma \Sigma^T = \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_r^2 \\ & & & 0 \end{bmatrix}_{D \times D} \quad \sigma_1 \geq \sigma_2 \geq \dots$$

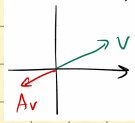
thus. $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ maximize the var $= \frac{1}{N} b^T \Sigma \Sigma^T b = \frac{1}{N} \sigma_1^2$

$u = U b =$ first col of $U =$ eigenvector with largest eigenvalue of $\frac{1}{N} A A^T = S$
 $=$ eigenvector with largest eigenvalue of $\frac{1}{N} \sigma_1^2$
 pos semi def eigenvalue ≥ 0



PCA doesn't work

eigenvalue/vector: preserve direction $Av = \lambda v$.



only direction matters

can rescale $v \Rightarrow A(\alpha v) = \lambda(\alpha v) \quad \forall \alpha \neq 0$

terminology: characteristic polynomial: $\det(A - \lambda I)$

spectrum: all roots of characteristic polynomial. can be repeated/complex

λ_i has algebraic multiplicity of m_i ; λ_i repeat m_i times

geometric multiplicity of λ_i : # linearly independent eigenvectors associated with the same eigenvalue λ_i
 $\dim(N(A - \lambda_i I))$

geometric multiplicity $\leq$ algebraic multiplicity

Example

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\det(A - \lambda I) = (3 - \lambda)(1 - \lambda) + 1 = (\lambda - 2)^2$$

$\lambda = 2$ is a repeated eigenvalue with algebraic multiplicity of 2

$$\begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 2 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{cases} 3v_1 - v_2 = 2v_1 \\ v_1 + v_2 = 2v_2 \end{cases} \Rightarrow v_1 = v_2 \Rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$0 = (A - 2I)x = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} x$$

$$\dim(N(A - 2I)) = 1$$

$\Rightarrow$ geometric multiplicity = 1 < 2

Example

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\pi_A(\lambda) = \det(A - \lambda I) = (2 - \lambda)(2 - \lambda)$$

$\lambda = 2$ is a root, repeated twice

$\hookrightarrow$ this is an eigenvalue with algebraic multiplicity $k = 2$

Find the eigenvectors:

$$Av = \lambda v$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 2 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Note any $v \in \mathbb{R}^2$ works.

Q: What is the dimension of the span of such v 's? $A: 2$

$$A - \lambda I = A - 2I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\dim N(A - \lambda I) = 2$$

This is an example of an eigenvalue with geometric multiplicity $m = 2$

Diagonalization

suppose $A \in \mathbb{R}^{n \times n}$ has n linearly independent eigenvectors $v_1, \dots, v_n$

$$\text{let } T = [v_1 \dots v_n]$$

$$AT = [\lambda_1 v_1 \dots \lambda_n v_n] = T \cdot \underbrace{\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}}_D$$

T is full rank, invertible

$$\Rightarrow A = TDT^{-1}$$

A is diagonalizable $\Leftrightarrow$ geometric multiplicity = algebraic multiplicity

LEC 14 Jordan canonical form
matrix exponential

$$\text{rotation matrix } A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\lambda_1, v_1 = i, \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$\lambda_2, v_2 = -i, \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

the eigenvector directions are preserved by A in complex plane

A is a rotation in real plane

for symmetric matrix A , if $\lambda_i \neq \lambda_j$ then $v_i \perp v_j$

for matrix A , if $\lambda_1 \neq \lambda_2 \neq \dots \neq \lambda_k$, then $v_1, v_2, \dots, v_k$ are linearly independent

Diagonalization $A = TDT^{-1}$

$T = [v_1 \dots v_n]$, $\{v_i\}$ are linearly independent eigenvectors

$$D = \text{diag}(\lambda_1, \dots, \lambda_n)$$

if eigenvalues of A are distinct, $\Rightarrow$ linearly independent eigenvectors $\Leftrightarrow A$ is diagonalizable

$$\Leftrightarrow \text{e.g. } \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad \lambda = 2/2$$

$$v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ still independent}$$

Jordan Canonical Form

A is not diagonalizable. we can still write $A = T J T^{-1}$

$$J = \begin{bmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_q \end{bmatrix} \quad \text{Jordan Canonical Form}$$

$\underbrace{\quad}_{k_1} \quad \underbrace{\quad}_{k_2} \quad \dots \quad \underbrace{\quad}_{k_q}$

$$J_i = \begin{bmatrix} \lambda_i & 1 & 0 & \dots & 0 \\ 0 & \lambda_i & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_i & 1 \\ 0 & \dots & 0 & 0 & \lambda_i \end{bmatrix} \quad k_1 + \dots + k_q = n$$

$$\Pi_A(\lambda) = \prod_{j=1}^q (\lambda - \lambda_j)^{k_j}$$

$$A^k = T D^k T^{-1}$$

Another application in differential equation:

$$\frac{dx}{dt} = Ax$$

If $x(t_0) = x_0$, then $x(t) = e^{(t-t_0)A} x_0$

because $\frac{d}{dt}(e^{tA}) = A e^{tA}$

matrix exponential

from calculus $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

the same holds for matrix $e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k \leftarrow \text{matrix exponential}$

$$= I + A + \frac{1}{2!} A^2 + \frac{1}{3!} A^3 + \dots$$

$$e^{tA} = \sum_{k=0}^{\infty} \frac{1}{k!} t^k A^k$$

Properties of the matrix exponential

$$① e^0 = I$$

$$② (e^A)^T = (I + A + \frac{1}{2!} A^2 + \dots)^T$$

$$= I + A^T + \frac{1}{2!} (A^T)^2 + \dots$$

$$= e^{A^T}$$

$$③ e^{(t+\tau)A} = e^{tA} \cdot e^{\tau A}$$

why? $e^{tA} e^{\tau A} = \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k \sum_{l=0}^{\infty} \frac{\tau^l}{l!} A^l$

$$e^{(t+\tau)A} = \sum_{n=0}^{\infty} \frac{(t+\tau)^n}{n!} A^n \quad \text{Binomial Theorem}$$

$$= \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{n!}{k!(n-k)!} t^k \tau^{n-k} \frac{1}{n!} A^n$$

let $l = n-k$

$$= \sum_{l=0}^{\infty} \sum_{k=0}^{\infty} \frac{1}{k!l!} t^k \tau^l A^{k+l}$$

$$= e^{tA} e^{\tau A}$$

$$④ \text{ If } AB=BA \text{ (A and B commute)}$$

$$\text{then } e^{t(A+B)} = e^{tA} e^{tB} = e^{tB} e^{tA}$$

(left as HW to check)

$$⑤ \frac{d}{dt}(e^{tA}) = \frac{d}{dt} \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k = \sum_{k=0}^{\infty} \frac{k t^{k-1}}{k!} A^k$$

$$= \sum_{k=1}^{\infty} \frac{t^{k-1}}{(k-1)!} A^k = A e^{tA} = e^{tA} A$$

Diagonalization of matrix exponential

$$\text{Suppose } A = T D T^{-1}$$

$$\text{Then } e^{tA} = \sum_{k=0}^{\infty} \frac{1}{k!} t^k (T D T^{-1})^k$$

$$= \sum_{k=0}^{\infty} \frac{1}{k!} t^k T D^k T^{-1}$$

$$= T \left(\sum_{k=0}^{\infty} \frac{t^k}{k!} D^k \right) T^{-1}$$

$$e^{tA} = T e^{tD} T^{-1}$$

$$\text{Note } D^k = \text{diag}(\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k)$$

$$\Rightarrow e^{tD} = \text{diag}(e^{t\lambda_1}, \dots, e^{t\lambda_n})$$

Moreover, if $T^{-1} = T^T$, then

$$e^{tA} = \sum_{i=1}^n e^{t\lambda_i} v_i v_i^T$$

symmetric matrix

spectral theorem

non-negative/positive
matrix

Perron-Frobenius Theorem

Markov matrix

Random walk

Symmetric matrix application

- Covariance matrices $\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(x_i - \bar{x})^T$
- positive definite / semidefinite matrices
 - $x^T A x > 0$ $\forall x \neq 0$
 - $x^T A x \geq 0$ $\forall x$
- Hessians $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$
- used in function minimization via Newton's method
- Quadratic forms $q(x) = x^T Q x$
e.g. $p(x, y) = x^2 + 4xy + 3y^2$
→ Matrix form $P(x, y) = [x, y] \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

Spectrum Theorem

For symmetric & real matrix: (i) eigenvalues are real

(ii) there exist an orthonormal & real basis of eigenvectors

$$A = Q \Lambda Q^T$$

non-negative/positive matrix

Consider any square matrix A

- A is nonnegative if all its entries ≥ 0 .
- A is positive if all its entries > 0 .

Q: If A and B are nonnegative, then:

- Is $A+B$ nonnegative? **Yes**
- Is $A \cdot B$ nonnegative? **Yes**
- Is A^{-1} nonnegative? **No**

counterexample $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

Perron-Frobenius Theorem

Perron-Frobenius Theorem: For any positive matrix A, then there exists a real, positive eigenvalue λ_* with multiplicity 1 such that

- 1) all other eigenvalues λ_i satisfy $|\lambda_i| < \lambda_*$
- 2) $A v_* = \lambda_* v_*$, where v_* is also positive, i.e. all entries $(v_*)_i > 0$.

Extension to Non-negative matrix?

- λ_* is nonnegative but could have multiplicity > 1
- v_* is nonnegative

(Think of a nonnegative matrix as the limit of a sequence of positive matrices)

$$\lim_{\epsilon \rightarrow 0^+} \begin{bmatrix} \epsilon & 1 \\ \epsilon & \epsilon \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ Eigenvalues are $\lambda_1 = 4$, $\lambda_2 = -1$ Associated eigenvector $v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$$A v_1 = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \end{bmatrix} = 4 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Suppose A is positive and diagonalizable then let $\lambda_1 = \lambda_*$ and observe

$$A = T \Lambda T^{-1}$$

$$A^k = T \Lambda^k T^{-1} = T \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} T^{-1}$$

$$= \lambda_1^k T \begin{bmatrix} 1 & & \\ & (\frac{\lambda_2}{\lambda_1})^k & \\ & & \ddots & \\ & & & (\frac{\lambda_n}{\lambda_1})^k \end{bmatrix} T^{-1}$$

$$\xrightarrow{k \rightarrow \infty} \lambda_1^k T \begin{bmatrix} 1 & & \\ & 0 & \\ & & \ddots & \\ & & & 0 \end{bmatrix} T^{-1}$$

Dominated by λ_1^k

Markov matrix / Stochastic matrix

a nonnegative matrix with an additional property that the sum of each row is 1

(Can also consider sum of each column! column stochastic vs row stochastic matrices)

Used to describe Markov chains

Example:



$$\begin{matrix} & a & b \\ a & \begin{bmatrix} 1/3 & 2/3 \end{bmatrix} \\ b & \begin{bmatrix} 1/2 & 1/2 \end{bmatrix} \end{matrix}$$

$$A_{ij} = \Pr(i \rightarrow j)$$

state described by a vector of probabilities

$$P_t = [P_t(a) \ P_t(b)]$$

$$P_{t+1}(j) = (P_t)_i = \sum_i P_t(i) \Pr(i \rightarrow j) = (P_t A)_j$$

$$A = \begin{bmatrix} 1/3 & 1/3 \\ 1/2 & 1/2 \end{bmatrix}$$

a row-stochastic matrix

$$A v = \lambda v$$

All rows sum to 1 $\Rightarrow (\lambda, v) = (1, [1])$
is an e-value/e-vector pair

$$\begin{bmatrix} 1/3 & 1/3 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Note that A and A^T have the same eigenvalues but different eigenvectors

$$A^T = \begin{bmatrix} 1/3 & 1/2 \\ 1/3 & 1/2 \end{bmatrix}$$

$$A^T v = 1 \cdot v$$

$$v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

• Stationary state: $P_* = [3/7 \ 4/7]$

$$P_* A = P_*$$

For A positive (or irreducible and aperiodic) the Markov chain converges to the stationary state.

$$P_t = P_0 A^t \xrightarrow{t \rightarrow \infty} P_*$$

Eigenvalues of Markov matrix $|\lambda| \leq 1$

Why? Suppose A is row-stochastic:

Example @ $n=3$: $Av = \lambda v$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \\ \lambda v_3 \end{bmatrix}$$

Consider the entry in v with the largest in absolute value. Suppose it's v_2 .

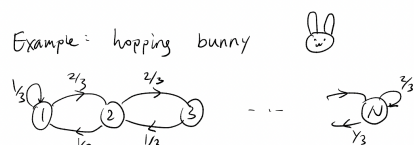
Then $|v_2| \geq |v_1|, |v_2| \geq |v_3|$

$$\begin{aligned} |\lambda v_2| &= |a_{21} v_1 + a_{22} v_2 + a_{23} v_3| \\ &\leq a_{21} |v_1| + a_{22} |v_2| + a_{23} |v_3| \\ &\leq a_{21} |v_2| + a_{22} |v_2| + a_{23} |v_2| \\ &\leq |v_2| \end{aligned}$$

$\Rightarrow |\lambda| \leq 1$

Application: Random Walk

Example: hopping bunny



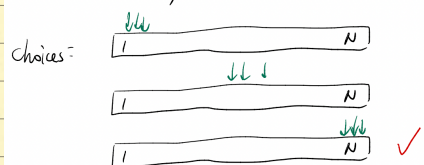
At each step, the bunny flips a biased coin

$\Pr(H) = 1/3, \Pr(T) = 2/3$

If H, jump left

If T, jump right

Q: After a long random walk, where is the bunny likely to end up?



To find the stationary state

$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ \vdots \\ P_n \end{bmatrix} = \begin{bmatrix} 1/2 & 1/3 & 0 & \dots & 0 \\ 2/3 & 0 & 1/3 & 0 & \dots & 0 \\ 0 & 2/3 & 0 & 1/3 & 0 & \dots & 0 \\ & & & 2/3 & 0 & 1/3 \\ & & & & 2/3 & 2/3 \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ \vdots \\ P_n \end{bmatrix}$$

$P = A P$

A is column-stochastic (square, nonnegative every column sums to 1).

p is the eigenvector of A with eigenvalue 1

Solve for P_k :

$P_1 = 1/2 P_1 + 1/3 P_2$

$2/3 P_1 = 1/3 P_2$

$2P_1 = P_2$

Next $P_2 = 2/3 P_1 + 1/3 P_3 = 1/3 P_2 + 1/3 P_3$

$2/3 P_2 = 1/3 P_3$

$2P_2 = P_3$

Recursive expression

$P_k = \frac{2}{3} P_{k-1} + \frac{1}{3} P_{k+1} \quad k=2, 3, \dots, N-1$

Suppose $2P_{k-1} = P_k$

$\Rightarrow P_k = \frac{1}{3} P_k + \frac{1}{3} P_{k+1} \Rightarrow P_{k+1} = 2P_k \quad \checkmark$

Example @ $N=10$

State k	1	2	3	4	5	6	7	8	9	10
$\frac{P_k}{P_1}$	1	2	4	8	16	32	64	128	256	512

$\sum_k P_k = 1 \Rightarrow P_1 \sum_k \frac{P_k}{P_1} = 1$

$1023 P_1 = 1$

$P_1 = \frac{1}{1023}$

nilpotent. For $A^{p+1} = 0_{nn}$ at some p

$$e^{tN} = I + tN + \dots + \frac{t^{p-1}}{(p-1)!} N^{p-1}$$

e.g. $N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $A = -I + N$. what is e^{tA}

I commutes with every matrix

$e^{tA} = e^{-t} \cdot e^{tN}$

$N^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is nilpotent

$e^{tN} = I + tN$

state $i \rightarrow j$. A_{ij} . column sum to 1

stationary. $Px = x$
eigenvalue 1

Q1: Given $Ax = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has no solution

$Ax = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ has 1 solution

A is $m \times n$ with rank r .

What can you say about (m, n, r) ?

$$m = 3, \quad m > n, r$$

$$1 \text{ soln} \Rightarrow N(A) = \{0\} \Rightarrow r = n$$

$$m = 3 > r = n$$

What would be an example A ?

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Q2: For any A in Q1, True or False?

1. $\det(A^T A) = \det(A A^T)$

False. Not generally true for rectangular A

Note: if A, B are square

$$\det(AB) = \det(BA) = \det(A) \det(B)$$

2. $A^T A$ is invertible

True, as $r = n$ (full column rank)

3. $A A^T$ is positive definite

False. $A A^T$ is positive semidefinite
(always true for any A)

Q3: How many solutions y to $A^T y = c$?

$$\dim N(A^T) = m - r > 0$$